

Exercise 1: Mocking and Stubbing

The image displays three sequential screenshots of an IDE (likely IntelliJ IDEA) showing the configuration and code for a Spring Boot application named 'mockitoexample'.

Top Screenshot: pom.xml

The Explorer view on the left shows the project structure. The main editor displays the `pom.xml` file. The content is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>mockitoexample</artifactId>
    <version>0.0.1</version>
    <packaging>jar</packaging>
    <name>mockitoexample</name>
    <description>mockitoexample</description>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter</artifactId>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>org.mockito</groupId>
            <artifactId>mockito-core</artifactId>
            <scope>test</scope>
        </dependency>
    </dependencies>
    <build>
        <plugins>
            <plugin>
                <groupId>org.springframework.boot</groupId>
                <artifactId>spring-boot-maven-plugin</artifactId>
            </plugin>
        </plugins>
    </build>
</project>
```

Middle Screenshot: MyService.java

The Explorer view on the left shows the project structure. The main editor displays the `MyService.java` file. The content is as follows:

```
package com.example.mockitoexample;

public class MyService {

    private ExternalApi api;

    public MyService(ExternalApi api) {
        this.api = api;
    }

    public String fetchData() {
        return api.getData();
    }
}
```

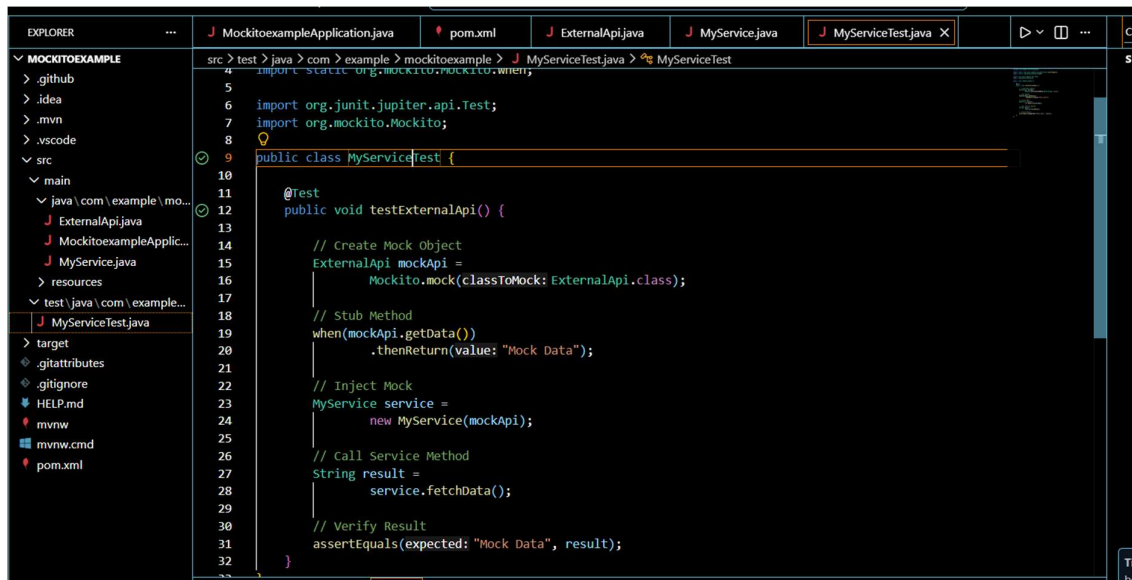
Bottom Screenshot: ExternalApi.java

The Explorer view on the left shows the project structure. The main editor displays the `ExternalApi.java` file. The content is as follows:

```
package com.example.mockitoexample;

public interface ExternalApi {

    String getData();
}
```



```
src > test > java > com > example > mockitoexample > J MyServiceTest.java > MyServiceTest
4 import static org.mockito.Mockito.when;
5
6 import org.junit.jupiter.api.Test;
7 import org.mockito.Mockito;
8
9 public class MyServiceTest {
10
11     @Test
12     public void testExternalApi() {
13
14         // Create Mock Object
15         ExternalApi mockApi =
16             Mockito.mock(classToMock: ExternalApi.class);
17
18         // Stub Method
19         when(mockApi.getData())
20             .thenReturn(value: "Mock Data");
21
22         // Inject Mock
23         MyService service =
24             new MyService(mockApi);
25
26         // Call Service Method
27         String result =
28             service.fetchData();
29
30         // Verify Result
31         assertEquals(expected: "Mock Data", result);
32     }
33 }
```

▼ **TERMINAL**

```
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ mockitoexample ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.mockitoexample.MyServiceTest
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader
classes because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.240 s -- in
com.example.mockitoexample.MyServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.863 s
[INFO] Finished at: 2026-06-25T20:12:19+05:30
[INFO] -----
❖ PS D:\mockitoexample>
```

Exercise 2: Verifying Interactions

```
src > test > java > com > example > mockitoexample > J MyServiceTest.java > {} com.example.mockitoexample

1 package com.example.mockitoexample;
2
3 import static org.mockito.Mockito.verify;
4
5 import org.junit.jupiter.api.Test;
6 import org.mockito.Mockito;
7
8 public class MyServiceTest {
9
10     @Test
11     public void testVerifyInteraction() {
12
13         // Create Mock Object
14         ExternalApi mockApi =
15             Mockito.mock(classToMock: ExternalApi.class);
16
17         // Inject Mock
18         MyService service =
19             new MyService(mockApi);
20
21         // Call Method
22         service.fetchData();
23
24         // Verify Interaction
25         verify(mockApi).getData();
26     }
27 }
```

com.example.mockitoexample.MyServiceTest

```
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.996 s
[INFO] Finished at: 2026-06-25T20:19:50+05:30
```